Key algorithms shaping the post-quantum cryptography landscape

| Algorithm | Features | Common use cases |
|---|---|---|
| Lattice-based cryptography algorithms | Built on the mathematical difficulty of lattice problems, these algorithms are designed to be resistant to both classical and quantum cyber attacks. They offer strong security guarantees and can be efficiently implemented, making them one of the most promising approaches for future cryptographic standards. | Securing digital communications, protecting confidential data, and serving as the foundation for advanced encryption and digital signature schemes in quantum-resilient systems. |
| Shor's algorithm | Unlike the others, Shor's algorithm is a quantum algorithm that can efficiently factor large integers and solve discrete logarithms. Its primary significance lies in its ability to break widely-used classical encryption techniques, such as RSA and ECC, if run on a sufficiently powerful quantum computer. | Primarily a threat model for existing cryptographic systems, Shor's algorithm is not used for securing data but serves as the catalyst for the development and adoption of quantum-resistant cryptography. |
| CRYSTALS-Kyber algorithm | This is a lattice-based key encapsulation mechanism that utilises the Learning With Errors (LWE) problem, making it highly resistant to quantum attacks. It is known for efficient key exchange and strong security even in the presence of quantum adversaries. | Ideal for establishing secure shared secrets over insecure channels, such as in encrypted messaging platforms and secure online transactions, ensuring that eavesdroppers cannot derive the exchanged keys. |
| CRYSTALS-Dilithium algorithm | Also founded on lattice-based constructs, Dilithium is a digital signature scheme relying on Module-LWE and Module-SIS problems. It offers a blend of robust security and practical performance, enabling swift signature generation and verification. | Widely used for authenticating software updates, verifying transactions, and ensuring the integrity and origin of digital communications, especially in environments where resistance to quantum attacks is paramount. |

Lattice-based cryptography stands out for its flexibility and strong quantum resistance, making it a cornerstone for next-generation encryption and signature mechanisms. In contrast, Shor's algorithm represents a pivotal breakthrough in quantum computing, serving as a reminder of the vulnerabilities in current cryptographic standards and driving the urgent transition to post-quantum solutions. CRYSTALS-Kyber offers a practical and secure method for exchanging keys, ensuring confidentiality in the face of both classical and quantum threats. Meanwhile, CRYSTALS-Dilithium guarantees authenticity and non-repudiation for digital signatures, addressing the growing need for trust in software distribution and electronic transactions.

These algorithms are being actively standardised and integrated into security protocols, providing a path for organisations to transition away from vulnerable systems and embrace quantum-resilient architectures.

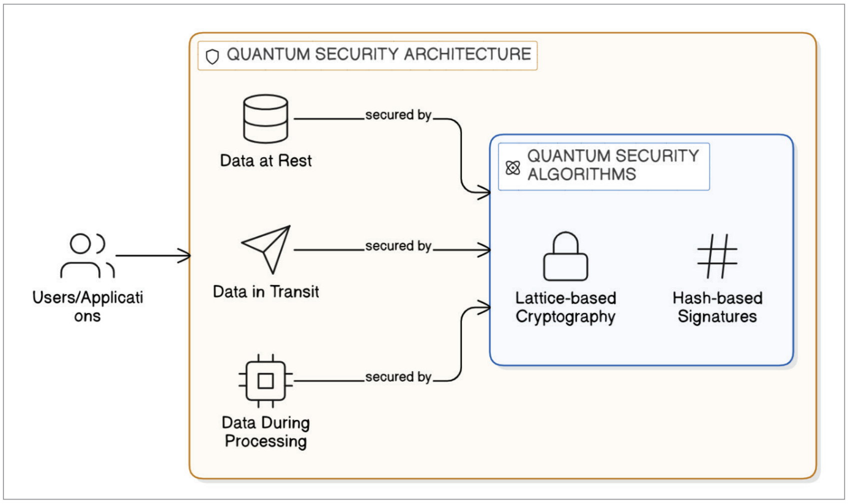


Figure 1: Quantum security reference architecture

## Post-quantum readiness assessment

As quantum threats loom on the horizon, assessing organisational readiness for post-quantum security becomes imperative. A thorough readiness assessment involves reviewing existing cryptographic assets, identifying vulnerable systems, and estimating the exposure to quantum-enabled attacks. This evaluation extends to strategic planning, resource allocation, and workforce training, ensuring that organisations are equipped to